Solid State

BITSAT · Nonofficial practice
Attempt: 555f398b-1b39-4e01-b301-901b5a253473
Started: 2026-10-17T12:20:20.830324+00:00
Content version: 1 · Key version: 1
Score version: original

Score: 2 / 150 · Percentage: 1.33% (not a percentile)
Correct: 1 · Incorrect: 1 · Skipped: 48
Dropped: 0 · Bonus-to-all: 0
Attempt rate: 4.0% · Accuracy: 50.0%
Accuracy = correct / attempted; dropped and bonus items excluded.
Penalty marks: 1 · Elapsed: 20 seconds
Rules: +3 correct, -1 wrong, 0 unanswered
Numerical: Single-correct MCQ: select A, B, C or D. No numeric-entry questions.

Complete record: all 50 questions in original delivery order.
Skipped questions and supplied answer keys are included. No worked solutions were supplied.
Downloaded copies are outside Tejas access controls.
Attribution: TEJAS200 · Owner-supplied original practice material

Frozen profile: e953b4ba9ab9316ad72a0e82c5bbc67290a84e5da1debde674a64df0b7e24ff1

Question 1 · Chemistry

Your response: B | Accepted answer(s): A
Marks: -1 | Incorrect | Reason: incorrect_response

Question

01. An fcc element has atomic molar mass $60\,\mathrm{g\,mol^{-1}}$ and edge $400\,\mathrm{pm}$. Taking $N_A = 6.0 \times 10^{23}\,\mathrm{mol^{-1}}$, density in $\mathrm{g\,cm^{-3}}$ is

(A) 6.25
(B) 3.125
(C) 25
(D) 12.5

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 22

Supplied answer key (no worked solution)

01	**A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 2 · Chemistry

Your response: C | Accepted answer(s): C
Marks: 3 | Correct | Reason: matched_accepted_key

Question

02. A metal changes from bcc to fcc while retaining the same atomic radius. Ratio of new to old cell edge is

(A) 1.4142136 (B) 2

(C) 1.2247449 (D) 0.81649658

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 22

Supplied answer key (no worked solution)

02	**C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 3 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

03. In an fcc anion cell, each tetrahedral site at $(1/4, 1/4, 1/4)$ has distance from the corner anion equal to

(A) $a/4$ (B) $\sqrt{3}a/4$

(C) $a/2$ (D) $\sqrt{2}a/4$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 22

Supplied answer key (no worked solution)

03 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 4 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

04. A crystal has $Fe_{0.95}O$ composition with iron only as Fe^{2+} and Fe^{3+}. Per mole of oxide ions, moles of Fe^{3+} are

(A) 0.1 (B) 0.2

(C) 0.05 (D) 0.15

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 22

Supplied answer key (no worked solution)

04 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 5 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

05. An fcc array of monovalent anions has half its octahedral sites occupied by cations. The cation charge required for neutrality is

(A) 4 (B) 1

(C) 3 (D) 2

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 22

Supplied answer key (no worked solution)

05 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 6 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

06. An fcc array of divalent anions has all tetrahedral sites occupied by identical cations. Cation charge is

(A) 2 (B) 3

(C) 4 (D) 1

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 23

Supplied answer key (no worked solution)

06 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 7 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

07. An fcc element has atomic radius r. If its atoms per cell are counted correctly but $a = 2r$ is incorrectly used, the computed density is how many times the true density?

(A) 1.4142136 (B) 4

(C) 2.8284271 (D) 2

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 23

Supplied answer key (no worked solution)

07 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 8 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

08. A crystal of $M_{0.90}O$ contains only M^{2+} and M^{3+}. What fraction of metal ions is trivalent?

(A) 0.22222222 (B) 0.11111111

(C) 0.1 (D) 0.2

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 23

Supplied answer key (no worked solution)

08 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 9 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

09. An fcc crystal has 4 atoms per cell, each with 12 nearest neighbours. Number of distinct nearest-neighbour contacts assigned per cell is

(A) 36 (B) 12

(C) 24 (D) 48

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 23

Supplied answer key (no worked solution)

09 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 10 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

10. In a zinc-blende-type AB crystal, anions are fcc and cations occupy half tetrahedral holes. The number of nearest-neighbour A-B contacts per conventional cell is

(A) 8 (B) 16
(C) 4 (D) 32

11–40 / New middle-to-hard practice

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 23

Supplied answer key (no worked solution)

10 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 11 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

11. An fcc metal consists of mass-58 and mass-62 isotopes in equal numbers. Its cell edge is 400 pm and density 5.625 $\mathrm{g\,cm^{-3}}$. Only random metal-site vacancies occur. With $N_A = 6.00 \times 10^{23}$, the vacant-site fraction is

(A) $\frac{1}{10}$ (B) $\frac{1}{20}$
(C) $\frac{9}{10}$ (D) $\frac{1}{5}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 23

Supplied answer key (no worked solution)

11 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 12 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

12. An fcc metal has cell edge 400 pm. Its density is initially 6.00 $\mathrm{g\,cm^{-3}}$; annealing fills all vacancies and raises density to 6.25 without changing the edge. No other defects occur. Its atomic molar mass and original vacancy fraction are, respectively ($N_A = 6.00 \times 10^{23}$)

(A) 60, 0.0400 (B) 62.5, 0.0400

(C) 60, 0.0800 (D) 57.6, 0.0400

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 23

Supplied answer key (no worked solution)

12 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 13 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

13. An fcc array contains oxide ions. A chemical assay finds 0.800 mol metal ions per mole oxide, all either A^{2+} in octahedral holes or B^{3+} in tetrahedral holes. No other ions or defects occur. Fraction of tetrahedral holes empty is

(A) $\frac{1}{5}$ (B) $\frac{3}{5}$
(C) $\frac{2}{5}$ (D) $\frac{4}{5}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 23

Supplied answer key (no worked solution)

13 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 14 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

14. A close-packed array of monovalent X anions has monovalent A in octahedral holes and divalent B in tetrahedral holes only. An assay gives A:B ion-number ratio 2 : 1. With no other defects, the fraction of tetrahedral holes filled by B is

(A) $\frac{3}{8}$ (B) $\frac{1}{4}$
(C) $\frac{1}{2}$ (D) $\frac{1}{8}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 24

Supplied answer key (no worked solution)

14 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 15 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

15. In $M_{1-x}O$, M occurs only as M^{2+} and M^{3+}. One-fourth of the metal ions are trivalent. The metal-site vacancy fraction x, relative to ideal MO, is

(A) $\frac{1}{8}$ (B) $\frac{1}{9}$
(C) $\frac{2}{9}$ (D) $\frac{1}{4}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 24

Supplied answer key (no worked solution)

15 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 16 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

16. A nonstoichiometric oxide has formula $M_{0.96}O$. All metal ions are either $+2$ or $+3$. The ratio of divalent to trivalent metal ions is

(A) 12 (B) 11

(C) $\frac{1}{11}$ (D) 23

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 24

Supplied answer key (no worked solution)

16 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 17 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

17. An initially ideal 111.000 g $CaCl_2$ crystal is doped by replacing some Ca^{2+} with Na^{+}. Charge compensation occurs only by chloride vacancies. Total cell count is unchanged; final crystal mass is 110.475 g. Fraction of original chloride sites now vacant is (Ca=40, Na=23, Cl=35.5)

(A) $\frac{1}{400}$
(B) $\frac{1}{100}$
(C) $\frac{1}{50}$
(D) $\frac{1}{200}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 24

Supplied answer key (no worked solution)

17 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 18 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

18. An initially ideal 58.500 g NaCl crystal is doped with Ca^{2+} on sodium sites. Compensation occurs only through vacant sodium sites. With total anion-site count unchanged, final mass is 58.440 g. Fraction of original sodium sites still occupied by sodium is (Na=23, Ca=40, Cl=35.5)

(A) $\frac{99}{100}$
(B) $\frac{199}{200}$
(C) $\frac{49}{50}$
(D) $\frac{24}{25}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 24

Supplied answer key (no worked solution)

18 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 19 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

19. A metal changes from bcc to fcc with unchanged atomic radius and atomic mass. The final density divided by its initial density is

(A) $\frac{\sqrt{6}}{2}$ (B) 2

(C) $\frac{4\sqrt{6}}{9}$ (D) $\frac{3\sqrt{6}}{8}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 24

Supplied answer key (no worked solution)

19 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 20 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

20. For one metal, bcc and fcc forms have identical measured density and no vacancies. If a_f and a_b denote their cell edges, the ratio of the hard-sphere radii inferred for fcc and bcc is

(A) $\frac{2^{\frac{5}{6}}\sqrt{3}}{3}$

(B) $\frac{\sqrt{6}}{2}$

(C) $\sqrt[3]{2}$

(D) $\frac{2^{\frac{5}{6}}\sqrt{3}}{6}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 24

Supplied answer key (no worked solution)

20 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 21 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

21. A fully occupied cubic 1:1 salt MX is known to adopt either rock-salt or CsCl structure. Its edge is 400 pm; diffraction gives nearest unlike-ion distance $200\sqrt{3}$ pm. Density is 3.00 $\mathrm{g\,cm^{-3}}$. Formula molar mass is $(N_A = 6.00 \times 10^{23})$

(A) $\frac{1152}{5}$ (B) $\frac{576}{5}$

(C) $\frac{288}{5}$ (D) $\frac{144}{5}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 24

Supplied answer key (no worked solution)

21 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 22 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

22. A fully occupied cubic 1:1 salt is either rock-salt or CsCl. Its edge is 600 pm, nearest unlike-ion separation is 300 pm, and density is 2.00 $\mathrm{g\,cm^{-3}}$. Formula molar mass is ($N_A = 6.00 \times 10^{23}$)

(A) $\frac{324}{5}$
(B) $\frac{648}{5}$
(C) $\frac{1296}{5}$
(D) $\frac{162}{5}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 25

Supplied answer key (no worked solution)

22 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 23 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

23. A fluorite-type crystal has 4 metal ions and 8 anion sites per cell. Half its divalent anions are replaced by monovalent ions. No vacancies occur, and metal ions are only +2 or +4. The fraction of metal ions in the +4 state is

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$
(C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 25

Supplied answer key (no worked solution)

23 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 24 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

24. An antifluorite M_2X initially has monovalent M and divalent X. One-fourth of M ions are removed, with no anion vacancies. The remaining M ions become a mixture of +1 and +2 only. The fraction of remaining M ions that must be divalent is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$
(C) $\frac{1}{3}$ (D) $\frac{2}{3}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 25

Supplied answer key (no worked solution)

24 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 25 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

25. An fcc oxide-ion array has 80.0% of its octahedral holes occupied by metal ions. Only divalent and trivalent metal ions are present; no other defects occur. The fraction of occupied metal sites that are trivalent is

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$
(C) $\frac{2}{5}$ (D) $\frac{3}{5}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 25

Supplied answer key (no worked solution)

25 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 26 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

26. A close-packed anion array has all tetrahedral holes filled by monovalent A and one-fourth of octahedral holes filled by divalent B. Anions are either 2− or 3−, with no other defects. The fraction of anions that are 3− is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$
(C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 25

Supplied answer key (no worked solution)

26 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 27 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

27. A bcc crystal has randomly vacant sites but unchanged atomic mass and cell edge. Its density is 7/8 of the fully occupied value. Relative to the perfect crystal, the expected fraction of nearest-neighbour pairs with both endpoints occupied is

(A) $\frac{7}{8}$ (B) $\frac{49}{64}$
(C) $\frac{1}{8}$ (D) $\frac{3}{4}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 25

Supplied answer key (no worked solution)

27 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 28 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

28. A fully occupied fcc model crystal contains N atoms. Atoms A and B have masses 10 and 40 u, with mean atomic mass 30 u. Labels are independently distributed. The expected number of distinct unlike nearest-neighbour pairs is

(A) $16N/3$ (B) $4N/3$

(C) $4N$ (D) $8N/3$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 25

Supplied answer key (no worked solution)

28 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 29 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

29. A 1 : 1 ionic crystal develops a Schottky vacancy fraction 1/20 on each sublattice. Simultaneously, its unit-cell edge shrinks to 99% of the initial value. Its final-to-initial density ratio is

(A) $\frac{970299}{950000}$
(B) $\frac{100000}{107811}$
(C) $\frac{950000}{970299}$
(D) $\frac{19}{20}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 26

Supplied answer key (no worked solution)

29 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 30 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

30. An MX crystal initially has 10^{20} sites on each monovalent sublattice. Cell dimensions stay fixed. It develops Schottky pairs plus cation Frenkel pairs; density is 98.0% of its initial value and cation interstitials number 10^{18}. Total normal-site vacancies on both sublattices are

(A) 4000000000000000000 (B) 3000000000000000000

(C) 5000000000000000000 (D) 6000000000000000000

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 26

Supplied answer key (no worked solution)

30 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 31 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

31. A silver-halide crystal is heated reversibly. Its mass, cell volume and stoichiometry remain constant, but mobile silver interstitials and an equal number of silver vacancies appear. Which combination best follows?

(A) Anion vacancies only; electron conductivity is necessarily the only current

(B) Frenkel defects; silver content must decrease; density increases

(C) Schottky defects; density must decrease; conductivity cannot change

(D) Frenkel defects; ideal density unchanged; ionic conductivity may increase

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 26

Supplied answer key (no worked solution)

31	**D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 32 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

32. An MX crystal has fixed cell dimensions and develops only Schottky pairs and cation Frenkel pairs. Density falls to 97.0% of its initial value. Per 10^{20} original cation sites, there are 2×10^{18} cation interstitials. Fraction of all normal cation vacancies belonging to Frenkel pairs is

(A) $\frac{1}{5}$ (B) $\frac{3}{5}$

(C) $\frac{2}{5}$ (D) $\frac{2}{3}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 26

Supplied answer key (no worked solution)

32 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 33 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

33. A silicon sample contains 1.10×10^{16} fully ionised donor plus acceptor atoms. Intrinsic carriers are negligible, and the net mobile carriers are 5.00×10^{15} electrons. The donor fraction among all dopant atoms is

(A) $\frac{8}{11}$ (B) $\frac{1}{2}$

(C) $\frac{3}{11}$ (D) $\frac{5}{11}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 26

Supplied answer key (no worked solution)

33 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 34 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

34. A silicon crystal has 1.40×10^{16} donor plus acceptor atoms. At the stated temperature only half the donors and all acceptors are ionised. Intrinsic carriers are negligible; net mobile electrons number 1.00×10^{15}. The donor fraction among all dopants is

(A) $\frac{1}{2}$ (B) $\frac{6}{7}$
(C) $\frac{5}{7}$ (D) $\frac{2}{7}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 26

Supplied answer key (no worked solution)

34 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 35 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

35. An fcc anion cell has edge a. A void on the diagonal has position (x, x, x), $0 < x < 1/2$, and is equidistant from the corner anion $(0, 0, 0)$ and face-centre anions such as $(0, 1/2, 1/2)$. Its distance to the cell-centre octahedral void, divided by the anion nearest-neighbour distance, is

(A) $\frac{\sqrt{6}}{2}$ (B) $\frac{\sqrt{3}}{4}$

(C) $\frac{1}{2}$ (D) $\frac{\sqrt{6}}{4}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 26

Supplied answer key (no worked solution)

35 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 36 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

36. In an fcc anion lattice, the nearest-neighbour anion-centre distance is $200\sqrt{2}$ pm. A tetrahedral-hole centre is $100\sqrt{3}$ pm from each surrounding anion centre. A small sphere fits exactly, touching those anions. Its radius is

(A) $100(\sqrt{2}-1)$ pm
(B) $100(\sqrt{3}-\sqrt{2})$ pm
(C) $100(\sqrt{3}-1)$ pm
(D) $100\sqrt{3}$ pm

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 27

Supplied answer key (no worked solution)

36 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 37 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

37. A close-packed array contains oxide ions. One-sixth of its tetrahedral holes contain D^{3+}; some octahedral holes contain M^{2+}. No other defects or ions occur. Charge neutrality gives which simplest formula?

(A) $M_3D_2O_6$ (B) M_3DO_6

(C) $M_2D_3O_6$ (D) MDO_3

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 27

Supplied answer key (no worked solution)

37 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 38 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

38. An fcc array consists of divalent X anions. All octahedral holes contain monovalent A; some tetrahedral holes contain divalent B. With no other defects, the simplest formula and fraction of tetrahedral holes occupied are

(A) A_4BX_4; 1/8
(B) A_2BX_2; 1/4
(C) ABX; 1/2
(D) AB_2X_2; 1/2

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 27

Supplied answer key (no worked solution)

38 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 39 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

39. A mixed oxide–chloride solid contains only a divalent metal, oxide ions and chloride ions. A sample contains 0.300 mol anions in total and 1.60 g oxygen. Its cation-to-anion number ratio is (O=16)

(A) $\frac{2}{3}$ (B) 1

(C) $\frac{3}{4}$ (D) $\frac{1}{2}$

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 27

Supplied answer key (no worked solution)

39 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 40 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

40. A solid contains only M^{3+}, oxide and chloride ions. A sample has 0.500 mol anions; after complete dissolution its chloride gives 43.05 g AgCl quantitatively. The amount of metal ions is ($M(\mathrm{AgCl}) = 143.5$)

(A) 1/6 mol
(B) 1/5 mol
(C) 7/30 mol
(D) 3/10 mol

41–50 / Preserved hard practice

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 27

Supplied answer key (no worked solution)

40 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 41 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

41. In a hypothetical cubic oxide, O^{2-} forms an fcc array and M^{3+} occupies only octahedral holes, with just enough M ions for neutrality. The edge is 400 pm, M=52, O=16, and $N_A = 6.0 \times 10^{23}\,\mathrm{mol}^{-1}$. The ideal density in $\mathrm{g\,cm}^{-3}$ is

(A) 6.25
(B) 7.5
(C) 3.3333333
(D) 5.2777778

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 27

Supplied answer key (no worked solution)

41 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 42 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

42. A cubic ionic solid has edge 400 pm, density 5.0 g/ cm^3 and formula mass 96 g/mol. Using N_A $=6.0\times10^{23}$, number of formula units per cell is

(A) 2 (B) 3

(C) 1 (D) 4

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 27

Supplied answer key (no worked solution)

42 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 43 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

43. A metal crystallizes in fcc with atomic radius $100\sqrt{2}$ pm. Its density is 6.25 g/ cm^3. Taking N_A $=6.0\times10^{23}$, atomic molar mass in g/mol is

(A) 30 (B) 90

(C) 120 (D) 60

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 28

Supplied answer key (no worked solution)

43 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 44 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

44. A 6.976 g sample of $Fe_{0.96}O$ contains only Fe(II) and Fe(III). Use Fe=56, O=16. After acid dissolution, Fe(II) is selectively oxidized to Fe(III) by acidic permanganate. Required MnO_4^- amount, in mol, is

(A) 0.0192
(B) 0.0176
(C) 0.02
(D) 0.0088

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 28

Supplied answer key (no worked solution)

44 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 45 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

45. In a hypothetical cubic ionic crystal, X^{3-} occupies all corners and the body centre, while M^{+} occupies all edge centres and face centres. If atomic masses are M=23 and X=31, the empirical-formula mass in g/mol is

(A) 123 (B) 100

(C) 77 (D) 54

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 28

Supplied answer key (no worked solution)

45 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 46 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

46. An fcc array of Cl^- contains Na^+ in all octahedral sites. A small fraction 0.01 of Na^+ is replaced by Ca^{2+}. To preserve neutrality using only cation vacancies, the number of vacant Na sites per 10000 original Na sites is

(A) 200 (B) 50

(C) 1000 (D) 100

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 28

Supplied answer key (no worked solution)

46 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 47 · Chemistry

Your response: Unanswered | Accepted answer(s): A
Marks: 0 | Skipped | Reason: unanswered

Question

47. A stoichiometric 1: 1 ionic crystal loses 1.00% of each ion species by Schottky -defect formation. Its number of unit cells remains fixed, but heating also increases every cell edge by 1.00%. Relative to the original crystal, the density decreases by what percentage?

(A) 3.9115754 (B) 1

(C) 4 (D) 2.9409852

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 28

Supplied answer key (no worked solution)

47 **A**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 48 · Chemistry

Your response: Unanswered | Accepted answer(s): D
Marks: 0 | Skipped | Reason: unanswered

Question

48. A monatomic bcc metal has edge 300 pm. A 1.00 cm^3 specimen dissolves quantitatively as M^{2+} in excess HCl, with hydrogen as the only reduced product. Use $N_A = 6.0 \times 10^{23}\,mol^{-1}$. The hydrogen amount formed, in mol, is

(A) 0.061728395
(B) 0.012345679
(C) 0.24691358
(D) 0.12345679

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 28

Supplied answer key (no worked solution)

48 **D**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 49 · Chemistry

Your response: Unanswered | Accepted answer(s): B
Marks: 0 | Skipped | Reason: unanswered

Question

49. An ionic compound has 4 formula units per cell and molar mass 120 g/mol. Its edge shrinks by 20% at constant crystal composition and number of cells. Density ratio final/initial is

(A) 0.512
(B) 1.953125
(C) 1.5625
(D) 1.25

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 28

Supplied answer key (no worked solution)

49 **B**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30

Question 50 · Chemistry

Your response: Unanswered | Accepted answer(s): C
Marks: 0 | Skipped | Reason: unanswered

Question

50. A 6.640 g sample of $Fe_{0.90}O$ contains Fe(II) and Fe(III) only. Use Fe=56, O=16. After acid dissolution, 0.0100 mol permanganate oxidizes Fe(II) selectively to Fe(III). All permanganate is consumed. The final iron mixture is then reduced quantitatively to metal. Required electron amount, in mol, is

(A) 0.27
(B) 0.18
(C) 0.25
(D) 0.2

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 29

Supplied answer key (no worked solution)

50 **C**

Source 87d786f6-1b8e-562f-ae5b-98848890e211 · page 30